Version Control and Collaboration

Branching

What Is a Branch?

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Question What is the name of your current branch?

Creating a Branch

Commands

```
git branch feature
git checkout feature
```

- Work independently
- No effect on `main`

Branch Isolation

Commands

```
echo "Feature work" >> notes.txt
git add notes.txt
git commit -m "Add feature work"

git checkout main
cat notes.txt
```

Key Insight: Branches isolate changes

Do some commits in both branches and check log

```
git log --graph --oneline  --all --decorate
```

Merging Branches

Command

```
git merge feature
```

- Fast-forward merge
- Pointer movement, not file copying

Do some commits in both branches, merge and check log

```
git log --graph --oneline  --all --decorate
```

Merge Conflict

- Same line changed in two branches
- Git cannot guess intent

Resolution Steps

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- `git commit`

Local vs Remote Repositories

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- Local: complete history
- Remote: shared coordination point
- `origin` = default remote

Cloning a Repository

Command

```
git clone <repo-url>
```

- Copies full history
- Sets up remote tracking

Create a repository on github/gitlab and clone it.

Push and Pull

Commands

```
git push
git pull
```

- `push`: publish local commits
- `pull`: fetch + merge

Why Pull Requests?

- Code review

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- Continuous Integration
- Protect `main` branch

Typical Workflow

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1. Create issue
2. Create feature branch
3. Commit changes
4. Push branch
5. Open Pull Request
6. Review and merge

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2. Commits are snapshots
3. Staging enables clean history
4. Branches are cheap
5. Merges reconcile histories
6. PRs enable collaboration

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Steps:

1. Clone the repository:
   ```
   git clone https://github.com/user/project.git
   ```
2. Make changes locally and commit:
   ```
   git add .
   git commit -m "Your message"
   ```
3. Push changes to GitHub:
   ```
   git push origin main
   ```
4. Pull updates from GitHub (when needed):
   ```
   git pull origin main
   ```

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Steps:

1. Initialize repository locally (if not already done):

   ```
   git init
   ```

2. Add GitHub repository as a remote:

   ```
   git remote add origin https://github.com/user/project.git
   ```

3. Push local history to GitHub:

   ```
   git push -u origin main
   ```

Note: GitHub repository should be empty (no README or commits).